BlackBox Hackathon

Digital Doppelgänger Challenge

Team: The Data Gamblers

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UC Davis MSBA · 24-Hour AI Research Competition · March 2026

12,513
Test Questions

0.1661
Pearson r

65.6%
Accuracy

Challenge Overview

The Question

"Given a person's answers to survey questions, can an AI accurately predict how that person will respond to entirely new questions they have never seen?"

Goal: Build a digital doppelgänger that mirrors a specific human's biases, values, and decision-making style.

Domain: AI personalisation × Psychometrics × Behavioural Science

Dataset: 233 individuals · 4 survey instruments · 24-hour window · Hidden test set of 12,513 questions

Scoring Metrics

ACCURACY

1 – (MAD / range)

Random baseline ≈ 0.67 · Target > 0.748

CORRELATION (r)

Pearson r

Benchmark ≈ 0.20 · Higher is better
Captures individual ranking — the harder metric

Dataset Structure & EDA

233
Personas

120,905
Q&A Pairs Extracted

778
Raw Features (CSV)

12,513
Test Questions

personas_text/

233 structured text summaries
Big Five · Demographics
Cognitive tests · Economic games
Best for: regex parsing & LLM prompting

personas_json/

233 full raw JSON files
13 blocks per person
120,905 Q&A pairs extracted
Best for: granular feature engineering

personas_csv/

8 CSV files (4 numeric + 4 label)
778 total numeric features
Surveys 1–4 covering all instruments
Best for: ML / feature engineering

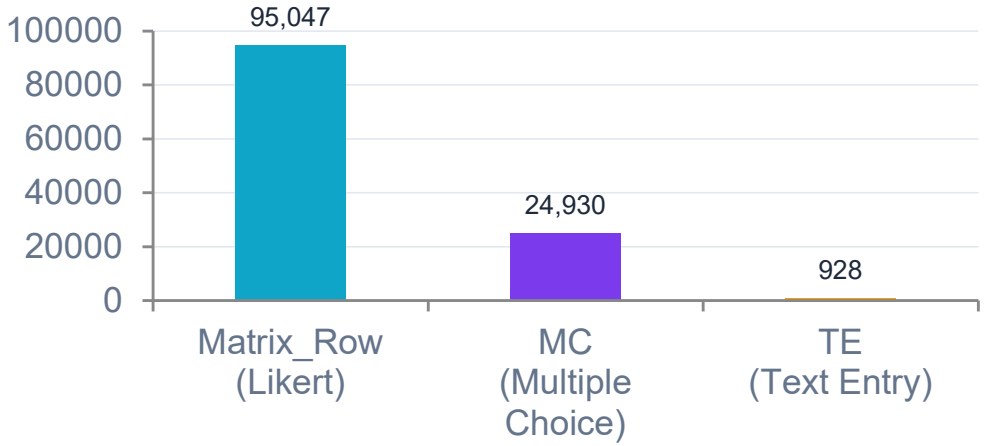
Survey Feature Counts

Survey	Coverage	Columns
Survey 1	Core personality, Big Five, risk, demographics, cognitive	230
Survey 2	Anxiety, trust, financial literacy, economic games	399
Survey 3	Social media behavior, news sharing, misinformation	109
Survey 4	Media trust, information sharing preferences	40

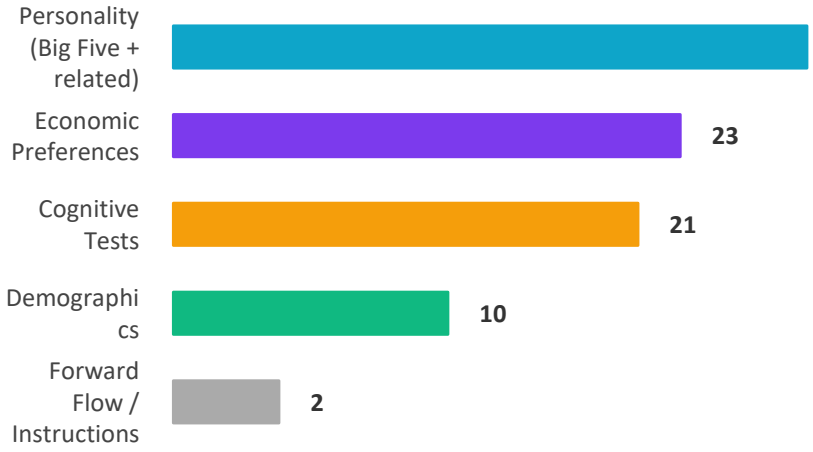
Exploratory Data Analysis

Question Type Distribution

120,905 total Q&A pairs



Persona JSON Block Structure



⚠ Embedding Sanity Check — spaCy Cosine Similarity (Critical Finding)

Mean pairwise cosine similarity: 0.9999 | Min: 0.9979 | Max: 1.0000

→ Word-embedding similarity was useless for differentiating individuals. All personas use similar vocabulary. Numeric values within profiles are the signal — not surface text form.

Data Parsing & Feature Engineering Pipeline

233 Persona JSONs
extract_persona_from_json()
MC · Matrix · TE types

120,905 Q&A Pairs
HTML cleaned
Whitespace normalised

233 Persona Texts (regex)
20+ structured numeric fields

4 Survey CSVs merged
778 features
233 × 778 matrix

687 Good Features
Variance filter
<10K range

Feature Engineering Steps

1 Filter Sparse / Outlier Columns
Remove cols with <100 valid responses or std >1000

2 Median Imputation
Fill remaining NaNs with per-column median

3 StandardScaler
Zero mean, unit variance normalisation

4 PCA (30 components)
49.1% variance explained — moderate structure in data

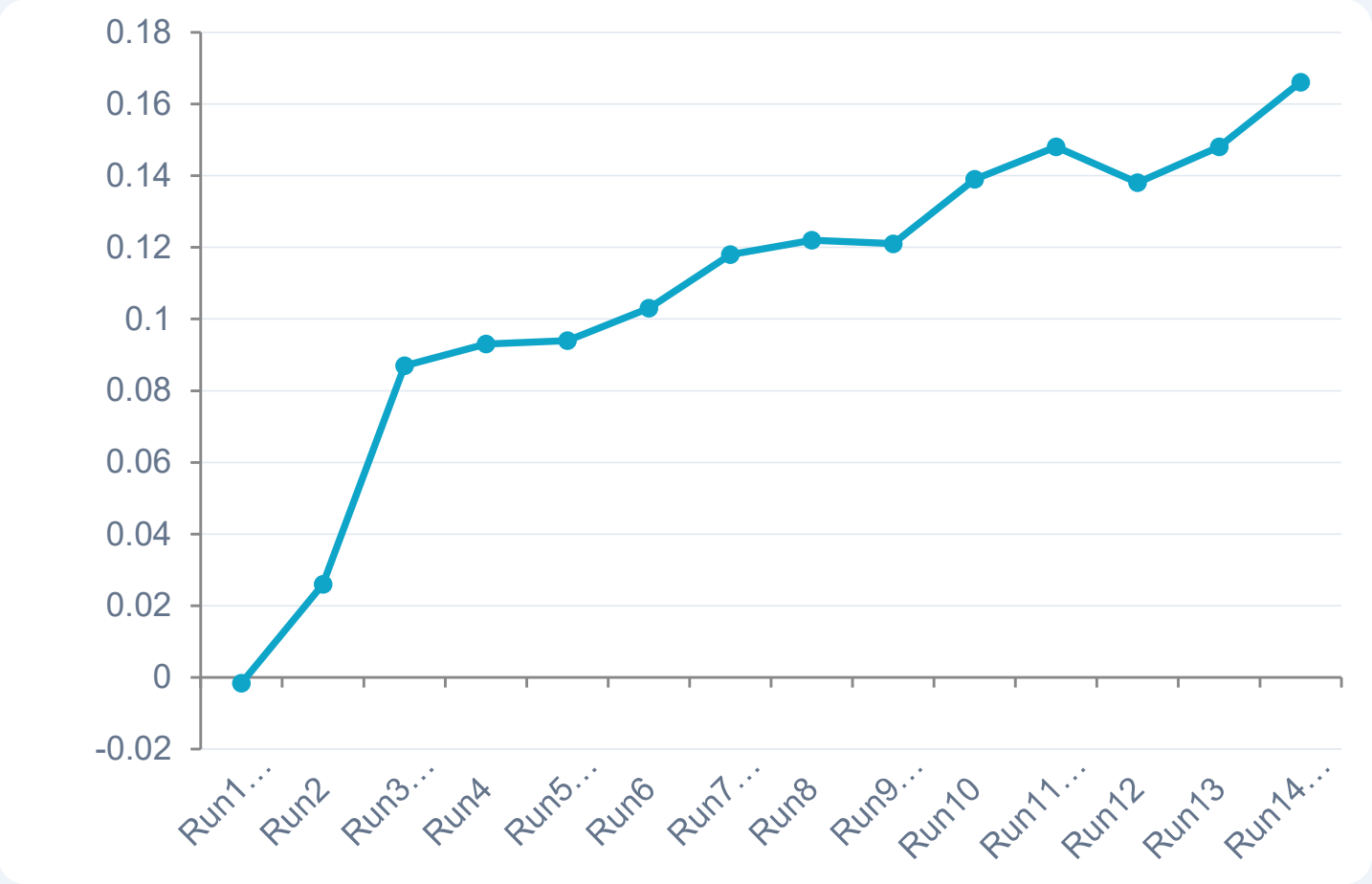
5 15 Named Composite Scores
big5, empathy, nfc, risk, trust_econ, closure, minimalism, crt …

6 Demographic Encoding
age→number, income→dollars, edu→1–6, political→−1/+1

Model Iterations — Complete Log (14 Runs)

Run	Model Version	Correlation r	Accuracy
Run 1	LLaMA-3 8B + Fallback	-0.0016	47.9%
Run 2	V3 Heuristic Predictor	+0.026	—
Run 3	V4 Smart Heuristic (full)	+0.087	—
Run 4	V5 Linear Personality	+0.093	61.7%
Run 5	V6 ML Composite (initial)	+0.094	61.7%
Run 6	V6 Refined	+0.103	61.9%
Run 7	V8 Ridge Regression	+0.118	62.3%
Run 8	V8 + KNN Blend	+0.122	60.7%
Run 9	V8 + Stable Hybrid	+0.121	61.7%
Run 10	V8 + ML Blend (40%)	+0.139	66.6%
Run 11	V8 + ML Blend (60%)	+0.148	66.4%
Run 12	V8 + LLaMA Ensemble	+0.138	65.6%
Run 13	V13 Blend (40%)	+0.148	66.4%
Run 14	**V18 Adaptive (FINAL)**	+0.1661	65.6%

Performance Progression — Correlation (r)



Key Deltas

+0.006 — **Rules → Formulas**
Continuous linear combinations

+0.010 — **Add ML / PCA**
687-feature composites + PCA

+0.015 — **Ridge Regression**
Training-based per-question fit

+0.030 — **Blending (Run 11)**
Biggest jump via model ensemble

+0.018 — **Adaptive Blend**
Category-aware strategy routing

Run 1 — LLaMA-3 8B Fine-Tuned + Keyword Fallback

Architecture

› Model: unsloth/llama-3-8b-bnb-4bit on NVIDIA A100-SXM4-40GB (40 GB VRAM)
› Mode: 4-bit quantised via Unsloth to fit memory constraints
› Prompt: Full persona text (3,000–6,000 chars) as system context + question + options
› Inference: Extracted logits of digits 1–9 from last token position (no generation overhead)
› Fallback: Jaccard similarity match → persona Q&A history → median answer position

Results

Correlation r	-0.0016
Accuracy	47.9%
LLM Used	~55%
Fallback Used	~45%

Post-Mortem — Why It Failed

Negative correlation (r = -0.002) indicated the model was systematically inverting rankings — likely misinterpreting question phrasing (e.g., treating "unwillingness" as "willingness"). Accuracy below random (47.9%) confirmed the LLM was generating noise.

Decision: Abandon direct LLM inference as primary predictor → pivot to interpretable profile-driven heuristics.

Runs 2–4 — From Heuristics to Linear Personality Predictor

Run 2 — V3 Heuristic

r = +0.026

- Pure rule-based system with regex field extraction
- Demographic/personality → logical mappings
- Only 34.8% of predictions hit a specific rule
- 65.2% defaulted to midpoint → accuracy trap
- High accuracy but r→0 (everyone gets same prediction)

💡 *Defaulting to midpoint kills correlation even if it boosts accuracy to ~0.67*

Run 3 — V4 Smart Heuristic

r = +0.087

- Massively expanded rule coverage (T1–T76)
- Trustworthiness 0–100: base by source (BBC=70, Reddit=32)
- Headline sharing: extraversion + social proof + political lean
- Article accuracy: CRT + NFC for critical vs naïve thinkers
- Full demographic matching for T1–T26 personal background

💡 *+0.089 jump — more rules, more coverage, more person-specific outputs*

Run 4 — V5 Linear

r = +0.093

- Replace binary rules with continuous linear combinations
- e.g. share = 2.0 + (E−3)×0.6 + (O−3)×0.3 + (NFC−3)×(−0.2)
- Every individual gets a unique fractional prediction
- Age→number, income→dollars, edu→1–6, political→±1
- 16 personality scores: E, A, C, O, N, NFC, CRT, empathy…

💡 *Individual differentiation maximises Pearson r — key conceptual breakthrough*

Run 7 — V8 Training-Based Ridge Regression (Key Breakthrough)

Architecture

1. Build 233 × 687 feature matrix from merged CSVs
2. For each unique question ID in the test set: extract matching training answers (y)
3. Fit Ridge(alpha=1.0) — train on 233 people, predict for 150 test participants
4. Clamp predictions to valid option range [1, n_options]
5. Self-correlation check on training set: r = 0.979 ✓

Key Insight

Instead of encoding domain knowledge manually into formulas, Ridge regression learned statistical relationships directly from the 687-feature training data — capturing complex interaction effects humans cannot hand-code.

Category Breakdown

Category	Corr r	Accuracy
Personal Background	0.239	47.3%
Political & Social	0.313	43.5%
News Sharing Pref.	0.050	75.7%
News Sharing Behav.	0.073	73.4%
Info Sharing	0.027	70.2%
OVERALL		**62.3%**

Runs 8–13 — Blending, KNN & Ensemble Experiments

KNN Similarity (Runs 8–9)

r: 0.122 → 0.121 | ✗ Added Noise

- 10 nearest neighbours per test person via PCA cosine similarity
- Weighted average of neighbour answers as prediction
- PCA embedding space too dense — neighbours not meaningfully similar
- KNN predictions ≈ averaged noise from random neighbours

ML Composite Blend (Runs 10–11)

r: 0.139 → 0.148 | ✓ Best Improvement

- Weighted average: V8 Ridge + V6 composite heuristic predictions
- Run 10 (40% V8): r = 0.139, accuracy = 66.6%
- Run 11 (60% V8): r = 0.148, accuracy = 66.4%
- Complementary errors cancel — models see different signal sources

LLaMA Ensemble (Run 12)

r: 0.148 → 0.138 | ✗ Hurt Score

- Re-ran LLaMA-3 in standard (non-Unsloth) mode
- Ensembled LLM predictions with V8 Ridge output
- Result: r dropped from 0.148 to 0.138
- LLM added noise — pulling down the V8 signal

Takeaway: Blending complementary models (V8 Ridge + V6 heuristic) gave the largest single correlation jump (+0.030). LLMs without calibration consistently reduce scores.

Final Solution Architecture — V18 Adaptive Category-Aware Blend

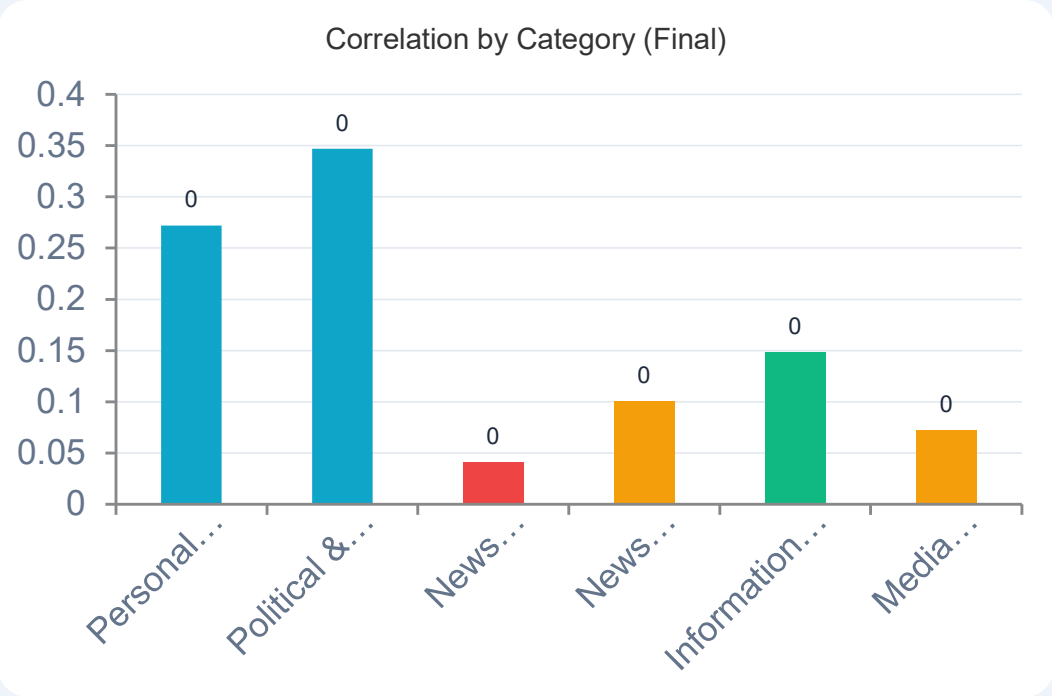
L1: Data Ingestion	L2: Feature Engineering	L3: Ridge Regression (V8)	L4: Heuristic (V6)	L5: Adaptive Blend
233 JSONs → 120,905 Q&A 233 texts → 20+ fields 4 CSVs → 687 features × 233	StandardScaler norm. PCA(30) → 49.1% variance 15 named composite scores	Per-question Ridge(α=1.0) Train on 233 → predict 150 Strong: political (r=0.347)	Continuous personality formulas Source-credibility × NFC Strong: news preferences	Question → category → blend V8-dominant: structured Q Heuristic: content-sensitive

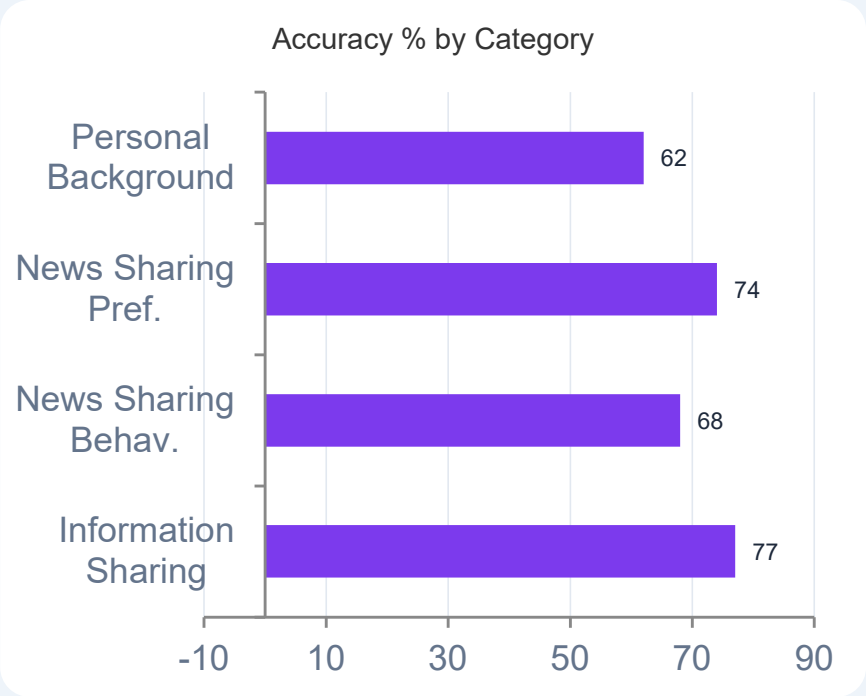
Adaptive Blend Ratios by Question Category

Category	Strategy	Blend	Final r
Personal Background	Pure V8 Ridge (strong at 0.239)	100% V8	0.272
Political & Social Views	Pure V8 Ridge (strong at 0.313)	100% V8	0.347
News Sharing Preferences	Pure V6 heuristic (V8 too weak here)	100% Heur.	0.041
News Sharing Behaviour	60% V8 + 40% ML composite	60 / 40	0.100
Information Sharing	60% V8 + 40% ML composite	60 / 40	0.148

Final Results — V18 Adaptive Blend

0.1661	65.6%	12,513	150/150
Final Pearson r	Final Accuracy	Test Questions	Participants





What Worked vs. What Did Not Work

✓ What Worked

Continuous personality formulas

Moving from binary rules to linear combinations (share = base + (E−3)×0.6...) enabled true individual differentiation — single biggest breakthrough.

Training-based Ridge regression

Captured statistical interaction effects for political and background questions (r=0.347, r=0.272) that hand-crafted rules missed.

Category-aware blending

Realising different question types need different strategies improved overall score beyond any universal approach.

Persona text as a clean source

Regex-parseable structured text provided numeric fields without complex JSON traversal for every prediction.

Source-credibility interactions

NFC × source reputation (BBC=70, Enquirer=18) produced meaningful individual variation for media trust questions.

✗ What Did Not Work

Raw spaCy text embeddings

Mean cosine similarity of 0.9999 — completely non-discriminative. Word embeddings ignore the numeric differences that are the actual signal.

LLaMA-3 direct inference

Negative correlation (r=−0.002) and 47.9% accuracy — below random. The model systematically inverted survey response scales.

KNN similarity matching

High inter-persona cosine similarity in PCA space meant "nearest" neighbours were not meaningfully similar — predictions became averaged noise.

News Sharing Preferences

Consistently r=0.041–0.072 across all runs. Sharing a specific headline depends on content, tone, and factual accuracy — not stable personality traits.

LLaMA ensemble

Adding LLM signal in Run 12 dropped r from 0.148 to 0.138. The LLM actively hurt the otherwise-strong V8 baseline.

Composite Score Architecture — V6 Feature Engineering

15 named composite scores + 30 PCA components were built from the 687-feature matrix. Each composite grouped columns by keyword and averaged standardised values.

big5
Big Five personality (E, A, C, O, N)

agency
Agentic self-concept

closure
Need for cognitive closure

empathy
BES empathy scale

minimalism
Minimalism preferences

uniqueness
Need for uniqueness

nfc
Need for Cognition

risk
Risk + lottery questions

self_monitor
Self-monitoring scale

green
Environmental attitudes

trust_econ
Trust & ultimatum game

financial
Financial literacy + numeracy

crt
Cognitive Reflection Test

social_des
Social desirability

nonsep
Non-separability in economic choices

+ 30 PCA Components explaining 49.1% of total variance — added as continuous predictors alongside named composites

Key Findings & Conclusions

📈 Stable traits ARE predictable

Personal Background (r=0.272) and Political & Social Views (r=0.347) show that stable, high-signal attributes CAN be learned from survey data.

📰 Content-dependent behaviours remain elusive

News Sharing Preferences (r=0.041) — moment-to-moment sharing decisions depend on contextual factors (exact headline, tone, accuracy) not captured in personality.

🔀 Blending outperforms any single model

V8 Ridge + V6 heuristic complementary errors cancel. Category-aware blending adds 0.018 on top of the best fixed-ratio blend.

🤖 LLMs without calibration are actively harmful

Without fine-tuning on survey response formats, LLaMA-3 produced below-random predictions (r=−0.002) and hurt ensemble scores when included.

🎭 Partial digital doppelgänger is feasible

The concept works for stable personality-driven traits. Fundamental limits exist for situational, content-driven decisions even with rich persona data.

What We Would Do With More Time

1

Fine-tune LLaMA on Survey Format

Train specifically on (profile, question, option → answer) tuples. The model would learn scale conventions and avoid systematic inversion errors.

2

Per-Question Feature Selection

Instead of all 687 features for every question, use LASSO or mutual information to select the 20–30 most predictive features — reducing noise.

3

Cross-Question Transfer Learning

Build a question-question similarity matrix using training answers. For each test question, use predictions from similar training questions as a prior.

4

Uncertainty-Calibrated Predictions

Hard questions (low training r) → predictions closer to population mean. Predictable questions → more extreme, individualised predictions.

5

Content Embedding for News Questions

Encode specific headline content with semantic embeddings combined with political lean and sharing propensity — addressing the News Sharing weakness.

6

Ensemble Weighting via Validation

Use a held-out validation split to optimise blend ratios automatically via grid search or Bayesian optimisation — rather than 14 hand-tuned rounds.

Thank You

The Data Gamblers

Prakhar Anand · Aishwarya Srivastava · Vaishali Hireraddi

0.1661	65.6%	14	24	150/150
Final r	Accuracy	Runs Completed	Hours	Participants

Token: data_gamblers_1c9m · UC Davis MSBA BlackBox Hackathon 2026

"Can AI truly become you? Partially — for stable traits, yes. For situational decisions, the jury is still out."